PRODUCT USER MANUAL

For OMI_CLIMATE_OFC_BALTIC_area_averaged_anomalies

Issue: 2.0

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_OFC_BALTIC_area_averaged_anomalies. It contains one dataset:

- omi_climate_ofc_baltic_area_averaged_anomalies

The OMI provides a timeseries of Baltic Sea area averaged fresh water content anomalies. The method for calculating the ocean freshwater content anomaly is based on the daily mean sea water salinity fields (S) derived from the Baltic Sea reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The total freshwater content is determined using the following formula:

$$OFC(t) = \iiint_V OFC(x, y, z, t)\, dx\, dy\, dz$$

The vertical integral is computed using the static cell vertical thicknesses sourced from the reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011 dataset cmems_mod_bal_phy_my_static, spanning from the sea surface to the ocean bottom. Spatial integration is performed over the Baltic Sea spatial domain, defined as the region between 9° - 31° E and 53° - 66° N.

We evaluate the uncertainty from the mean standard deviation of monthly mean OFC. The shaded area in the figure corresponds to the annual standard deviation of monthly mean OFC.

Linear trend (km3y-1) has been estimated from the annual OFC values with the uncertainty of 1.96-times standard error.

More details and full scientific evaluation can be found in the Copernicus Marine Service Ocean State Report 7 (Raudsepp et al., 2023).

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The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CLIMATE_OFC_BALTIC_area_averaged_anomalies
Geographical coverage	Baltic Sea
Variables	Ocean freshwater content anomaly
Available time series	From 1993 to Year-1
Temporal resolution	Yearly
Horizontal resolution	Integrated
Integration depth (for example)	0 – 300 m
Format	NetCDF4

b) Production System Description

The Baltic Sea area averaged ocean freshwater content anomalies have been calculated using the results from Copernicus Marine Service BALMFC Reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The vertical integral is computed using the static cell vertical thicknesses sourced from the reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011 dataset cmems_mod_bal_phy_my_static, spanning from the sea surface to the ocean bottom. Spatial averaging is performed over the Baltic Sea spatial domain, defined as the region between 9° - 31° E and 53° - 66° N.

c) Dataset details

omi_climate_ofc_baltic_area_averaged_anomalies contains the following variables:
ofc_balrean [km^3] long_name: Area Averaged Ocean Freshwater Content Anomaly from Baltic Sea Physical Reanalysis standard_name: sea_water_volume **ofc_balrean_lower_rmsd** [km^3] long_name: Lower Bound of Error of the Area Averaged Ocean Freshwater Content Anomaly from Baltic Sea Physical Reanalysis standard_name: sea_water_volume **ofc_balrean_upper_rmsd** [km^3] long_name: Upper Bound of Error of the Area Averaged Ocean Freshwater Content Anomaly from Baltic Sea Physical Reanalysis standard_name: sea_water_volume **time** [days since 1900-01-01 00:00:00] long_name: time

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

d) Nomenclature of original files

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-CLIMATE-OFC-area-averaged-anomalies)
Boyer, T., Levitus, S., Antonov, J., Locarnini, R., Mishonov, A., Garcia, H., Josey, S.A., 2007. Changes in freshwater content in the North Atlantic Ocean 1955–2006. Geophysical Research Letters, 34(16), L16603. Doi: 10.1029/2007GL030126

IOC, SCOR and IAPSO, 2010: The international thermodynamic equation of seawater - 2010: Calculation and use of thermodynamic properties. Intergovernmental Oceanographic Commission, Manuals and Guides No. 56, UNESCO (English), 196 pp. Available from http://www.TEOS-10.org (11.10.2021).

Urmas Raudsepp, Ilja Maljutenko, Amirhossein Barzandeh, Rivo Uiboupin, and Priidik Lagemaa, 2023. Baltic Sea freshwater content. 7th

edition of the Copernicus Ocean State Report (OSR7), Copernicus Publications, State Planet, 1-osr7, 7, https://doi.org/10.5194/sp-1-osr7-7-2023